Chapter 1
Matrices and Systems of Equations

1.5 Elementary Matrices

Objectives

- Elementary Matrices
- Triangular Matrices

Elementary matrices

- In this section, we view the process of solving a linear system in terms of matrix multiplications rather than row operations.
- Given an $m \times n$ linear system $A\mathbf{x} = b$, we can obtain an equivalent system by multiplying both sides of the equation a nonsingular $m \times m$ matrix:

$$MA\mathbf{x} = Mb$$

- Can transform the system $A\mathbf{x} = b$ to a simpler form that is easier to solve by multiplying a sequence of nonsingular matrices to the equation.

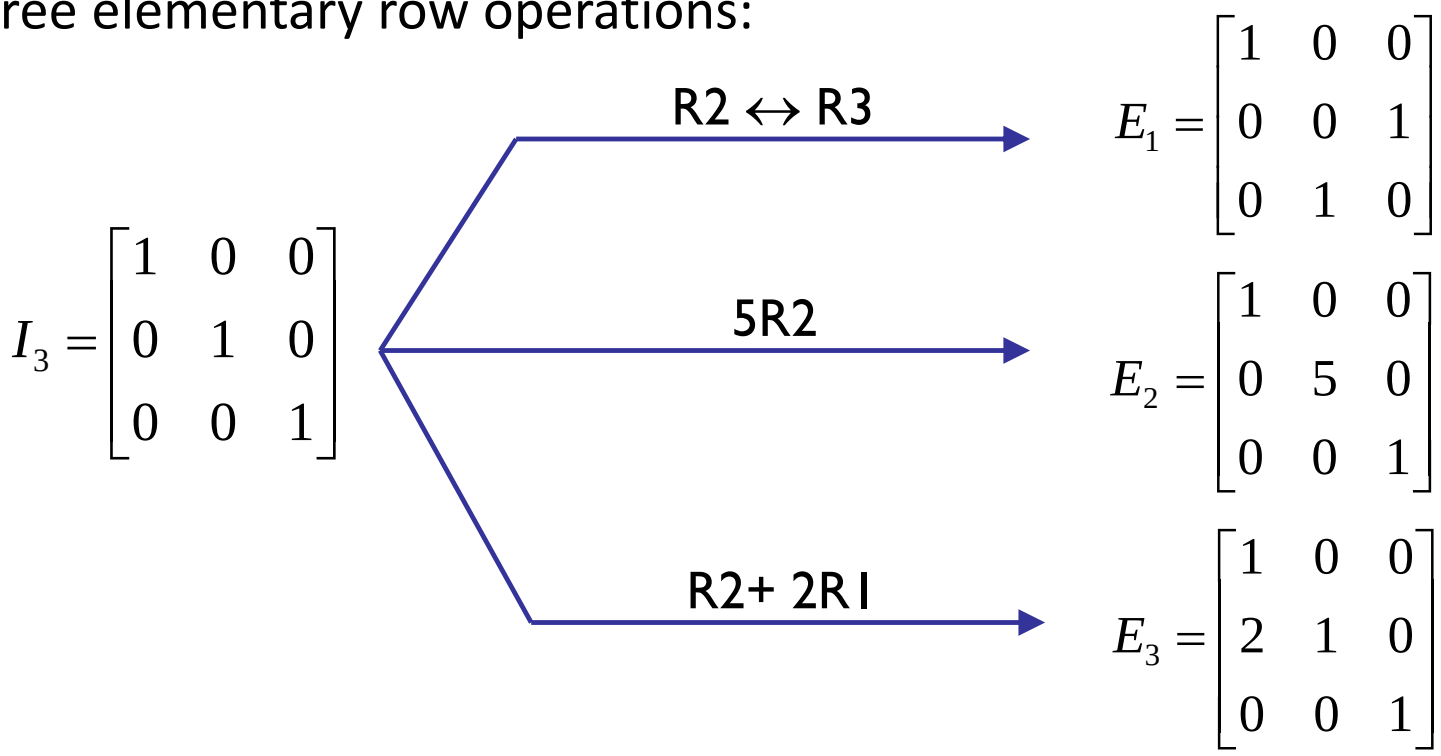
Elementary Matrices

Definition

An **elementary matrix** is one that can be obtained from the identity matrix I_n through a single elementary row operation.

Example 1

Three elementary row operations:



Elementary Matrices

- Type 1. A matrix obtained by interchanging two rows of I. E.g.,

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \qquad R_2 \leftrightarrow R_3$$

$$E_1 A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

$$AE_1 = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{13} & a_{12} \\ a_{21} & a_{23} & a_{22} \\ a_{31} & a_{33} & a_{32} \end{bmatrix}$$

Multiplying on the left by E1 interchanges rows R2 and R3 of A, and multiplication by E1 on the right interchanges columns C2 and C3 of A.

- Type 2. A matrix obtained by multiplying a row of I by a nonzero real number. E.g.,

$$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad k \times R_2, k \neq 0$$

$$E_2A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ ka_{21} & ka_{22} & ka_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$AE_2 = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & ka_{12} & a_{13} \\ a_{21} & ka_{22} & a_{23} \\ a_{31} & ka_{32} & a_{33} \end{bmatrix}.$$

Multiplying on the left by E2 multiplies R2 of A by k and multiplication by E2 on the right multiplies C2 of A by k.

- Type 3. A matrix obtained by adding a nonzero multiple of one row of I to another row, e.g.,

$$E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix} \qquad R_2 + k \times R_3, k \neq 0$$

$$E_3 A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} + ka_{31} & a_{22} + ka_{32} & a_{23} + ka_{33} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$AE_3 = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & ka_{12} + a_{13} \\ a_{21} & a_{22} & ka_{22} + a_{23} \\ a_{31} & a_{32} & ka_{32} + a_{33} \end{bmatrix}.$$

Multiplying on the left by E3 adds k times R3 of A to R2, and multiplication by E3 on the right adds k times C2 of A to C3.

- In general, suppose E is an $n \times n$ elementary matrix. We can think of E as being obtained from I by either a row operation or a column operation.
- If A is an $n \times r$ matrix, **pre-multiplying** A by E (i.e., EA) has the effect of performing that same row operation on A.
- If B is an $m \times n$ matrix, **post-multiplying** B by E (i.e., BE) is equivalent to performing that same column operation on B.

Elementary Matrices

elementary row operation ⇔
left multiply with an elementary matrix

$$A=\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

R2 ↔ R3

$$\begin{bmatrix} a & b & c \\ g & h & i \\ d & e & f \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} A = E_1 A$$

5R2

$$\begin{bmatrix} a & b & c \\ 5d & 5e & 5f \\ g & h & i \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix} A = E_2 A$$

R2+ 2R1

$$\begin{bmatrix} a & b & c \\ d+2a & e+2b & f+2c \\ g & h & i \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A = E_3 A$$

Theorem

If E is an elementary matrix, then E is nonsingular and E^{-1} is also an elementary matrix of the same type.

Proof

1. If E is an elementary matrix by interchanging the ith and j^{th}rows of I. Then E can be changed back to I by interchanging the same rows again. $EE = I$.
2. If E is an elementary matrix by multiplying the ith row with nonzero scaler α. Then E can be changed back to I by multiplying the same row with $\frac{1}{\alpha}$ (E).
3. If E is an elementary matrix by adding m times the ith row to j^{th} row. Then E can be changed back to I by subtracting m times the ith row from j^{th} row .

Theorem

If E is an elementary matrix, then E is nonsingular and E^{-1} is also an elementary matrix of the same type.

$$E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \ddots & 0 & \vdots \\ \vdots & 0 & \alpha & 0 \\ 0 & \dots & 0 & 1 \end{bmatrix}$$

$$E^{-1} = \begin{bmatrix} 1 & & & & & \\ & \ddots & & & & O \\ & & 1 & & & \\ & & & 1/\alpha & & \\ & & & & 1 & \\ & O & & & & \ddots \\ & & & & & & 1 \end{bmatrix} \quad i\text{th row}$$

$$E = \begin{bmatrix} 1 & & & & & & \\ \vdots & \ddots & & & & & O \\ 0 & \cdots & 1 & & & & \\ \vdots & & & \ddots & & & \\ 0 & \cdots & m & \cdots & 1 & & \\ \vdots & & & & & \ddots & \\ 0 & \cdots & 0 & \cdots & 0 & \cdots & 1 \end{bmatrix} \begin{matrix} \\ \\ i\text{th row} \\ \\ j\text{th row} \\ \\ \\ \end{matrix}$$

$$E^{-1} = \begin{bmatrix} 1 & & & & & & \\ \vdots & \ddots & & & & & O \\ 0 & \cdots & 1 & & & & \\ \vdots & & & \ddots & & & \\ 0 & \cdots & -m & \cdots & 1 & & \\ \vdots & & & & & \ddots & \\ 0 & \cdots & 0 & \cdots & 0 & \cdots & 1 \end{bmatrix}$$

Row equivalent

Theorem

A matrix B is row equivalent to A if there exists a finite sequence $E_1, E_2, \ldots, E_k$ of elementary matrices such that

$$B = E_k E_{k-1} \cdots E_1 A$$

Properties:

- If A is row equivalent to B, then B is row equivalent to A.
- If A is row equivalent to B, and B is row equivalent to C, then A is row equivalent to C.

Proof ? (exercise)

Theorem

If *A* and *B* are row equivalent matrices and *A* is invertible, then *B* is invertible.

Proof

If $A \sim \ldots \sim B$, then
$B = E_n \cdots E_2 E_1 A$ for some elementary matrices $E_n \cdots E_2$ and E_1.
So

$$B^{-1} = (E_n \cdots E_2 E_1 A)^{-1}$$
$$= A^{-1} E_1^{-1} E_2^{-1} \cdots E_n^{-1}$$ *since A and E_n are invertible*

Theorem

Equivalent Conditions for Nonsingularity

Let A be a $n \times n$ matrix. The following are equivalent.

(a) A is nonsingular.

(b) $A\mathbf{x}=0$ has only the trivial solution.

(c) A is row equivalent to I.

Corollary:

The system of n linear equations with n unknowns $A\mathbf{x} = b$ has a unique solution if and only if A is nonsingular.

- If A is nonsingular, then A is row equivalent to I.

  Therefore there exist elementary matrices

  $E_1, E_2 \ldots, E_k$ such that

$$E_k E_{k-1} \cdots E_1 A = I$$

- If we multiply on the right by A^{-1}, then

$$E_k E_{k-1} \cdots E_1 I = A^{-1}$$

- Therefore the same row operations that transform A into I, also transform I into A^{-1}. That is to say, the reduced row echelon form of the augmented matrix $[A|I]$ will be $[I\ |A^{-1}]$

- We can use this fact to find the inverse matrix!

Gauss-Jordan Elimination for finding the Inverse of a Matrix

Let A be an $n \times n$ matrix.

1. Adjoin the identity $n \times n$ matrix I_n to A to form the matrix $[A \mid I_n]$.
2. Compute the reduced echelon form of $[A \mid I_n]$.

   If the reduced echelon form is of the type $[\, I_n \mid B]$, then B is the inverse of A.

   If the reduced echelon form is not of the type $[\, I_n \mid B]$, i.e. the left-hand side $n \times n$ matrix is not I_n, then A has no inverse.

An $n \times n$ matrix A is invertible if and only if its reduced echelon form is I_n.

Example 2

Determine the inverse of the matrix $A = \begin{bmatrix} 1 & -1 & -2 \\ 2 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$

Solution

$$[A \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & 1 & 0 & 0 \\ 2 & -3 & -5 & 0 & 1 & 0 \\ -1 & 3 & 5 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[\text{R3}+\text{R1}]{\text{R2}+(-2)\text{R1}} \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & 1 & 0 & 0 \\ 0 & -1 & -1 & -2 & 1 & 0 \\ 0 & 2 & 3 & 1 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{(-1)\text{R2}} \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & 2 & 3 & 1 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[\text{R3}+(-2)\text{R2}]{\text{R1}+\text{R2}} \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 3 & -1 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & 0 & 1 & -3 & 2 & 1 \end{array}\right]$$

$$\xrightarrow[\text{R2}+(-1)\text{R3}]{\text{R1}+\text{R3}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 5 & -3 & -1 \\ 0 & 0 & 1 & -3 & 2 & 1 \end{array}\right]$$

Thus, $A^{-1} = \begin{bmatrix} 0 & 1 & 1 \\ 5 & -3 & -1 \\ -3 & 2 & 1 \end{bmatrix}$.

Example 3 (Using Example 2)

Solve the system of equations

$$x_1 - x_2 - 2x_3 = 1$$
$$2x_1 - 3x_2 - 5x_3 = 3$$
$$-x_1 + 3x_2 + 5x_3 = -2$$

Solution

This system can be written in the following matrix form:

$$\begin{bmatrix} 1 & -1 & -2 \\ 2 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$$

If the matrix of coefficients is invertible, the unique solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & -2 \\ 2 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$$

This inverse has already been found in Example 2. We get

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 5 & -3 & -1 \\ -3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

The unique solution is $x_1 = 1, x_2 = -2, x_3 = 1$.

Example 4

Determine the inverse of the following matrix, if it exist.

$$A = \begin{bmatrix} 1 & 1 & 5 \\ 1 & 2 & 7 \\ 2 & -1 & 4 \end{bmatrix}$$

Solution

$$[A : I_3] = \left[\begin{array}{ccc|ccc} 1 & 1 & 5 & 1 & 0 & 0 \\ 1 & 2 & 7 & 0 & 1 & 0 \\ 2 & -1 & 4 & 0 & 0 & 1 \end{array}\right]$$

$$\sim \begin{matrix} \\ R2+(-1)R1 \\ R3+(-2)R1 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 1 & 5 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & -3 & -6 & -2 & 0 & 1 \end{array}\right]$$

$$\sim \begin{matrix} \\ R1+(-1)R2 \\ R3+3R2 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 2 & -1 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 0 & -5 & 3 & 1 \end{array}\right]$$

There is no need to proceed further.

The reduced echelon form cannot have a one in the (3, 3) location.

The reduced echelon form cannot be of the form $[I_n : B]$.

Thus A^{-1} does not exist.

Triangular matrices

- A square $n \times n$ matrix $A = (a_{ij})$ is *upper triangular* if $a_{ij} = 0$ for $i > j$.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & \cdots & a_{1n} \\ 0 & a_{22} & a_{23} & \cdots & \cdots & a_{2n} \\ 0 & 0 & a_{33} & \cdots & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & & \vdots \\ \vdots & \vdots & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & a_{nn} \end{bmatrix}$$

- The products or sums of two upper triangular matrices are still upper triangular.
- Type III elementary matrices corresponding to "upper row – k*lower row" are upper triangular.

Triangular matrices

- It is *lower triangular* if $a_{ij} = 0$ for $i < j$.

$$A = \begin{bmatrix} a_{11} & 0 & 0 & \cdots & \cdots & 0 \\ a_{21} & a_{22} & 0 & \cdots & \cdots & 0 \\ a_{31} & a_{32} & a_{33} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & & \vdots \\ \vdots & \vdots & \vdots & & \ddots & 0 \\ a_{n1} & a_{n2} & a_{n3} & \cdots & \cdots & a_{nn} \end{bmatrix}$$

- The products or sums of two lower triangular matrices are still lower triangular.
- Type III elementary matrices corresponding to "lower row – k*upper row" are lower triangular.

Diagonal matrices

- *A diagonal matrix* is both upper and lower triangular, that is to say $a_{ij} = 0$ for $i \neq j$.

$$A = \begin{bmatrix} a_{11} & 0 & 0 & \cdots & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & \cdots & 0 \\ 0 & 0 & a_{33} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & & \vdots \\ \vdots & \vdots & \vdots & & \ddots & 0 \\ 0 & 0 & 0 & \cdots & \cdots & a_{nn} \end{bmatrix}$$

- A diagonal matrix can be written as a product of elementary matrices of type II. For example:

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

LU decomposition (optional)

- If an $n \times n$ matrix A can be reduced to strict upper triangular form U, using only row operation III (lower row minus multiple of upper rows), then $A = LU$, where L is lower triangular and U is upper triangular.
- Proof: If $E_k \dots E_2 E_1 A = U$ where E_i are lower triangular elementary matrices, and U is upper triangular, then

$$A = E_1^{-1} E_2^{-1} \dots E_k^{-1} U = LU$$

  Where $L = E_1^{-1} E_2^{-1} \dots E_k^{-1}$ is lower triangular.
- *Example: see book exercise 6*
- LU decomposition of large matrices is very important in engineering applications.

Summary

- 3 types of elementary matrices correspond to 3 types of row/column operations.
  - Multiplying elementary matrices on the left corresponds to row operations,
  - Multiplying elementary matrices on the right corresponds to column operations.
- Matrix A is nonsingular if and only if A is row equivalent to I_n.
  - To calculate the inverse matrix of a nonsingular A, write down $[A|I_n]$ and then row reduced it to $[I_n|A^{-1}]$.
- There are special matrices called upper/lower triangular matrices and diagonal matrices. Many matrices can be written as a lower triangular matrix times an upper triangular matrix.